

Exploring Disney Plus Content

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DATASCI 200: Section 010

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Background Overview

Our aim is to address the key question: **What content strategies contribute to higher ratings and awards for Disney Plus movies and series?** Our insights are tailored for **Disney Plus Content Strategists, Analysts, and Production Teams** who seek to understand the factors driving audience reception and critical acclaim. These insights can help refine content strategies to maximize viewer satisfaction and industry recognition. To break down the main question, we explored the following sub-categories:

1. IMDb Ratings Analysis

- o How do IMDb ratings differ between movies and series?
- o How do IMDb ratings vary by release year?
- o Do certain months have higher IMDb ratings for released content?
- o How does the runtime affect IMDb ratings?
- o How closely do critic ratings (Metascore) align with audience ratings (IMDb)?

2. Awards Analysis

- o How do the number of awards and nominations correlate with IMDb ratings?
- o Which Disney Plus movies or series have received the most awards, and how do their IMDb ratings compare?

3. Diversity and Genre Analysis

- o Do movies or series with more languages tend to have higher IMDb ratings?
- o Do movies or series with more country representations tend to have higher IMDb ratings?
- o Do animations receive higher IMDb ratings compared to other content types?

4. Sentiment and Box Office Analysis

- o How do movie plot sentiments align with IMDb ratings?
- o Which genres perform highly in the Box Office?

Data Sources

Primary Dataset: We used the Disney Plus Movies and TV Shows dataset sourced from Kaggle ([link to dataset](#)), containing detailed information about Disney Plus content such as titles, genres, release dates, IMDb ratings, Metascores, and awards. This dataset contains 992 rows and 19 columns and is available in our GitHub repository named: [disney_plus_shows.csv](#)

Supplementary Data: For additional context, we integrated Box Office Data from [Box Office Mojo](#).

Data Dictionary for Primary Dataset

Column Name	Description
imdb_id	Unique identifier for the title on the Internet Movie Database (IMDb).
title	Title of the movie or TV show.
plot	Brief plot summary of the movie or TV show.
type	Type of content (movie, series, or series episode).
rated	Content rating (e.g., G, PG, PG-13, R) or N/A if not available.
year	Year the movie or series was released.
released_at	Release date of the movie or series.
added_at	When the dataset was refreshed.
runtime	Duration of the movie or TV episode (e.g., 90 min, 1h 30 min).
genre	Genres associated with the content (e.g., Comedy, Drama).
director	Director(s) of the movie or series.
writer	Writer(s) or screenwriter(s) of the movie or series.
actors	Main actors and actresses featured in the movie or series.
language	Languages available for the movie or series (e.g., English, Spanish).
country	Country or countries where the movie or series was produced.
awards	Awards and nominations received by the movie or series.
metascore	Metascore rating from critics (out of 100).
imdb_rating	IMDb user rating (out of 10).
imdb_votes	Number of user votes on IMDb.

Data Dictionary for Supplementary Dataset

Column Name	Description
worldwide	Box Office earnings across all countries (revenue in billions).
title	Movie Titles.

Data Exploration and Cleaning

Primary Dataset Overview

The dataset contains 992 entries and 19 columns, detailing Disney Plus movies and series. The primary columns include IMDb ratings, plot summaries, genres, release dates, and awards/nominations. The initial data exploration revealed a few issues requiring cleaning and preprocessing to make the dataset suitable for analysis.

Key Observations:

- A number of columns contain missing values.
- Most columns are of the object data type, except for `metascore` and `imdb_rating`.
- Duplicate entries were checked, and 0 duplicates were found.

Handling Missing Data

Steps were taken to address missing data in the dataset. The `added_at` column was dropped since it only indicated when the data was added and did not provide meaningful insights. Rows with missing values in either `imdb_id` or `title` were also removed, as these identifiers are essential for accurate analysis. The `year` column was eliminated because the release year could be directly extracted from the `released_at` column. Additionally, rows with missing values in `imdb_rating`, `imdb_votes`, and `released_at` were dropped since these columns are crucial for ratings and time-based analysis. The proportion of missing data in these columns was relatively small, ranging from 1.68% to 2.24% of the dataset. In regards to the IMDb box office data, missing values in the `worldwide` column was handled by dropping filling empty rows with '0'. This column was also converted to numeric. Since the box office data only accounts for Disney Box office movies, the data frame is significantly smaller (225 rows).

Convert Data Types

To ensure the accuracy and consistency of the dataset, the data types of key columns were verified and converted as needed. The `released_at` column was converted to datetime format to support date-based analysis. The `metascore` column, which contained missing values, was converted to float after addressing these gaps. The `imdb_rating` column was confirmed to be a float, while the `imdb_votes` column was converted to integer to facilitate numerical analysis.

Feature Engineering

To enhance the dataset and simplify the analysis, new feature columns were created. The `released_at` column was used to generate two new columns: `release_month` and `release_year`, which help analyze trends by month and year. The `runtime` column, which contained durations in varied formats like "1h", "30 min", and "22 min", was standardized by creating a new column, `runtime_minutes`, representing the total runtime in minutes.

The `awards` column, which contained text entries such as "5 wins & 13 nominations," was transformed into a numerical column named `total_awards_nominations` to consolidate the total number of wins and nominations for easier analysis. For the `language` column, a new feature called `number_of_languages` was created to count the number of languages listed for each entry. Similarly, the `country` column was used to generate the `number_of_countries` column, capturing the total number of countries involved in the production.

The `genre` column, which listed multiple genres per entry, was utilized to create the `number_of_genres` column, indicating the total number of genres associated with each movie or series. Additionally, an `animation_category` column was derived from `genre` to classify entries as "Animation" if "Animation" was listed as one of the genres; all other entries were categorized as "Other."

Assumptions

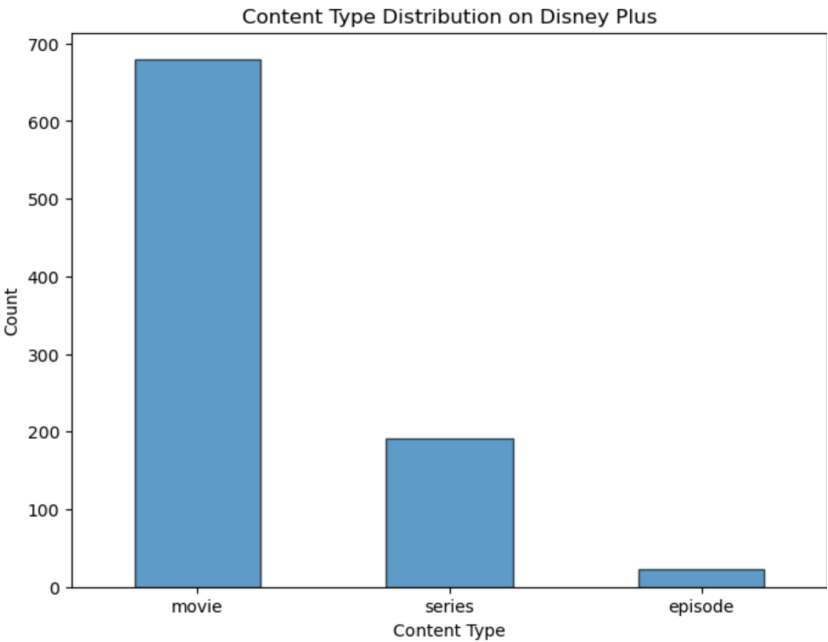
To guide our analysis and ensure clarity, the following assumptions were made:

- **Content Categorization:** All content not classified as Animation in the `genre` column is assumed to be Live Action.
- **Genre Classification:** The first genre listed in the `genre` column is assumed to be the primary genre for the content.
- **IMDb Ratings & Metascores:** IMDb ratings are assumed to accurately reflect audience perceptions and Metascores are assumed to accurately reflect critic's perceptions.
- **Multilingual Content:** The number of languages listed in the `language` column is assumed to reflect the content's accessibility and multilingual availability.
- **Awards and Nominations:** The total number of awards and nominations listed in the `awards` column is assumed to represent critical acclaim and recognition accurately.
- **Missing Data:** Missing values for languages, countries, or ratings are assumed to be due to data unavailability rather than inherent content limitations.

Exploratory Data Analysis

In this Exploratory Data Analysis (EDA), we aim to uncover insights that address our main question: What content strategies contribute to higher ratings and awards for Disney Plus movies and series? Using descriptive statistics and visualizations, we explored IMDb ratings, awards, diversity factors among other variables to identify key trends and relationships that can inform effective content strategies.

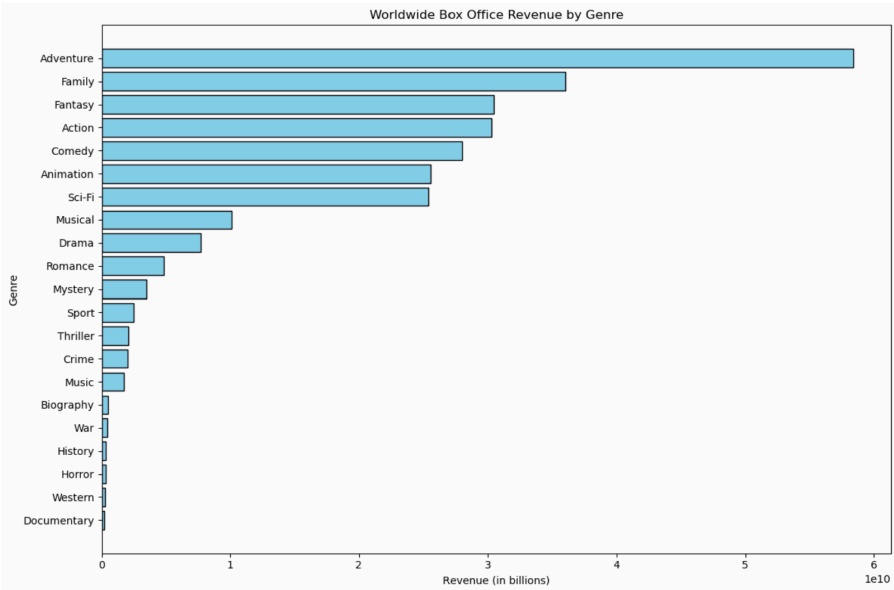
Content Type Distribution on Disney Plus



The distribution chart below highlights the platform's focus on providing a specific type of content to its audience. The most prominent content type is movies, with around **700 entries**, making them the largest category on Disney Plus.

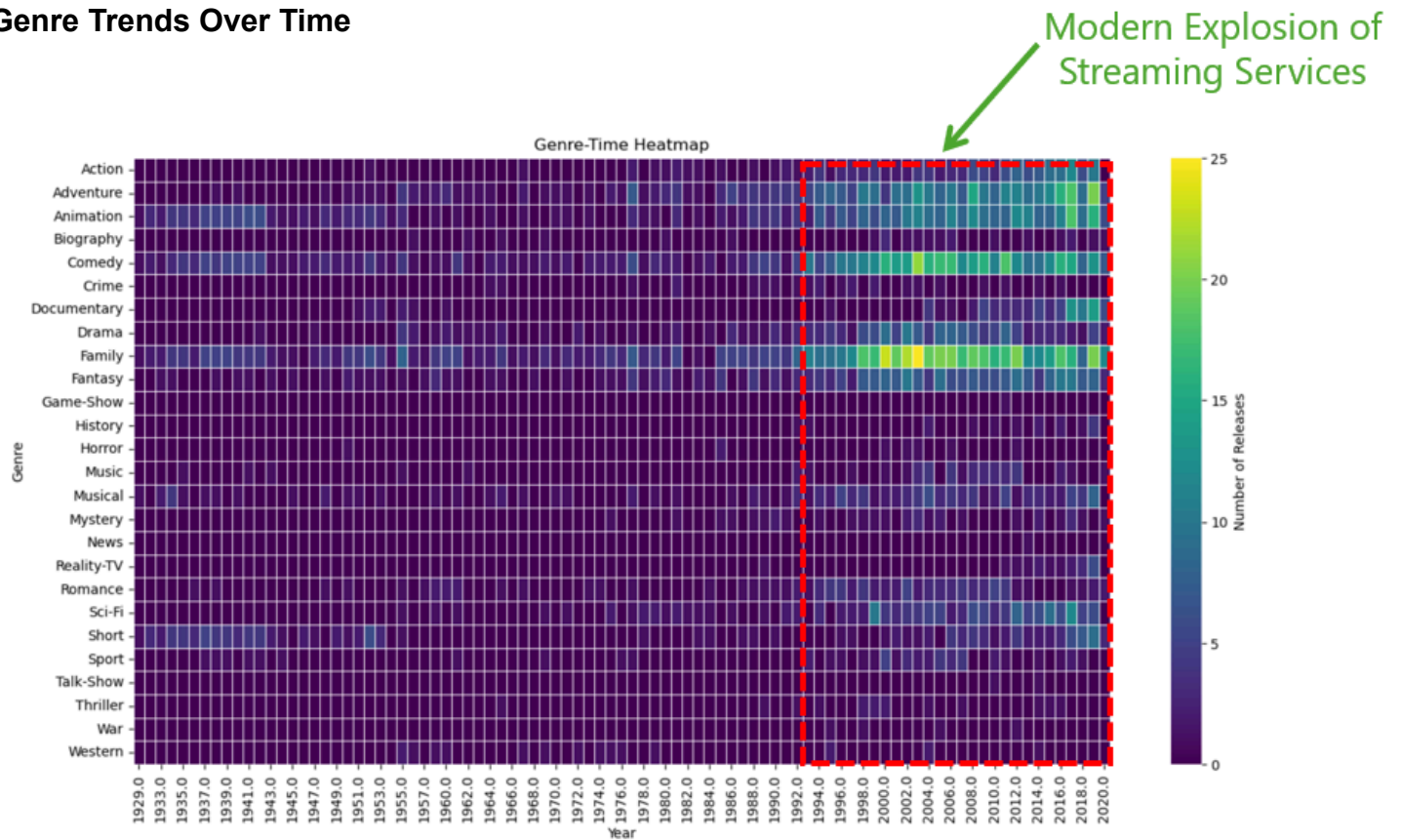
Genre Analysis

Which genres perform highly in the Box Office?



This inverted bar chart illustrates the worldwide box office revenue by different movie genres and highlights which genres dominate global box office earnings and which ones generate comparatively to lower ones. The “Adventure” genre leads by a much greater margin, generating the highest worldwide revenue. “Family”, “Action”, “Comedy”, and “Animation” are also among the highest generating genres.

Genre Trends Over Time



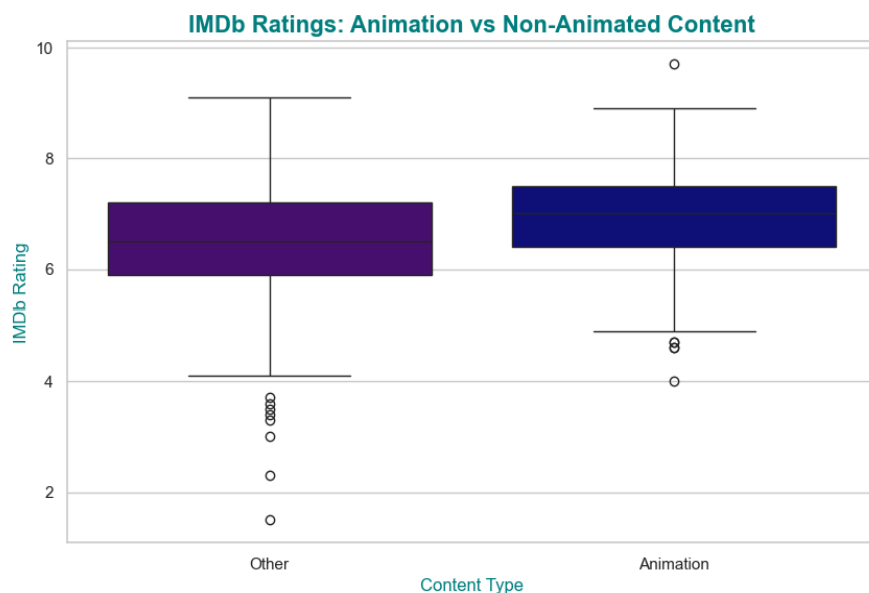
The heatmap visualizes the number of films released for each genre over time from 1928 to 2020 and allows us to see trends and patterns in genre popularity, in regard to movie releases, across different decades. The color intensity indicates the number of releases, with bright colors representing more releases. “Animation”, “Family”, and “Comedy” have consistently been dominant genres throughout the entire period. Gradual diversity in genres can also be seen. An explosion of movie releases can be seen after 1992, after “Line Internet Video” was developed.

Tree-map of Genre Distribution



The treemap illustrates the distribution of genres with the largest rectangle representing the “Family” genre. This reflects Disney Plus’ strong focus on family oriented programming, which is central to brand identity. Genres like “Comedy” and “Animation” also occupy a large amount of space, highlighting the platform’s commitment to entertainment and visually appealing content. Genres such as “Adventure” and “Fantasy” are also well represented, aligning with the popularity of franchises like Marvel and Star Wars.

Do animations receive higher IMDb ratings compared to other content types?



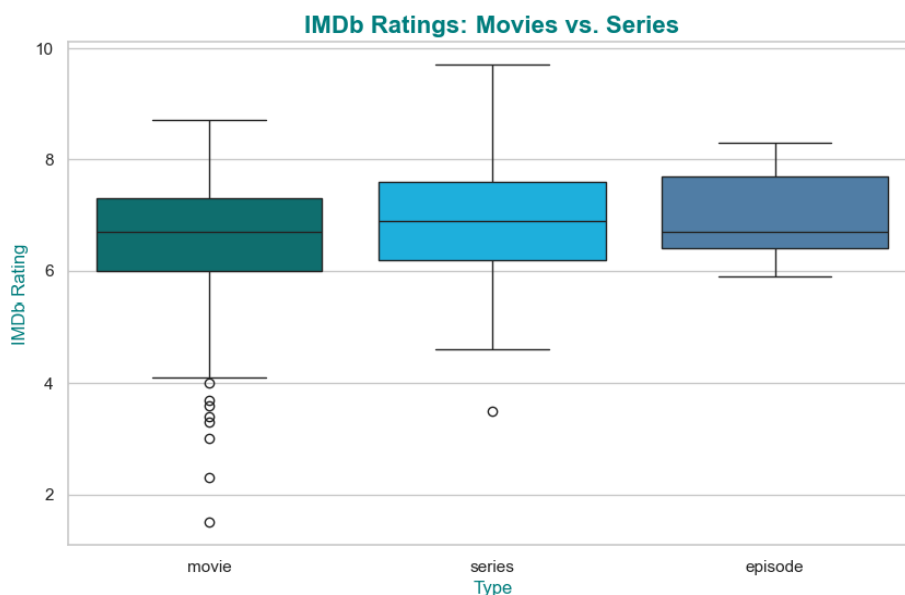
The box plot illustrates the relationship between IMDb ratings for Disney Plus animations and non-animations. Animations, comprising 362 titles, have an average IMDb rating of 6.92. The ratings for animations are tightly clustered, with a standard deviation of 0.86, and the middle 50% of ratings fall between 6.40 and 7.50. This indicates that most animated content is consistently well-received.

In contrast, non-animated content types, comprising 502 titles, have a lower average IMDb rating of 6.47. The ratings for non-animated content are more spread out, with a standard deviation of 1.09. The middle 50% of these ratings range between 5.90 and 7.20, reflecting greater variability in quality.

In summary, Disney Plus animations generally achieve higher and more consistent IMDb ratings compared to non-animated content. This suggests that Disney's animated offerings maintain a more reliable quality standard, reinforcing their long standing reputation for excellence in animation.

IMDb Rating Analysis

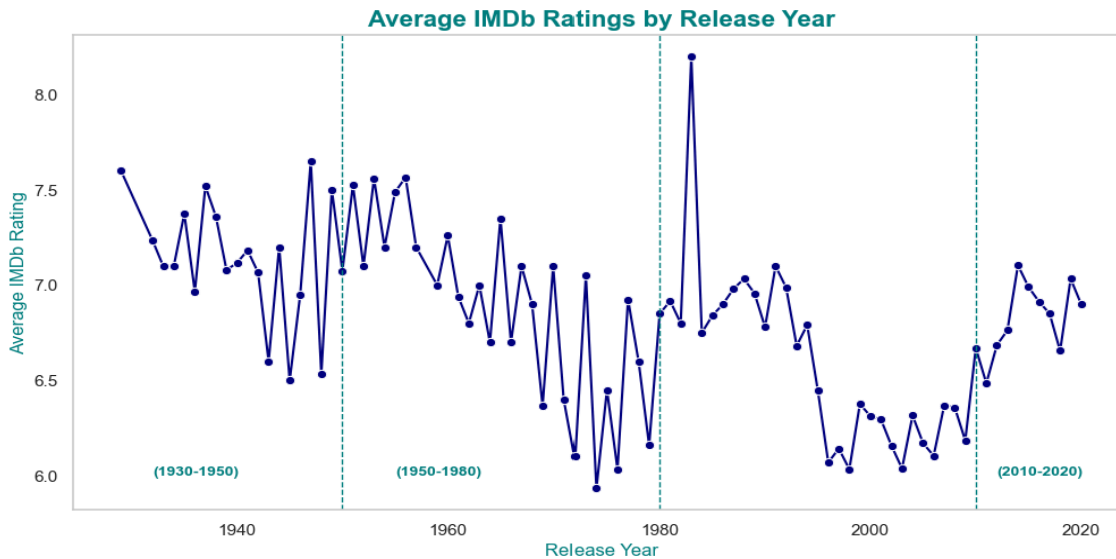
How do IMDb ratings differ between movies and series?



The box plot shows the distribution of IMDb ratings across movies, series, and episodes on Disney Plus. Movies have a median IMDb rating of around 6.5, with ratings spread widely and some outliers below 4, indicating varied quality. Series, on the other hand, have a slightly higher median IMDb rating of around 7.0, with fewer low outliers and a more consistent distribution. Episodes also have a median IMDb rating of around 6.5 but show a tighter spread, suggesting more consistent ratings compared to movies. In summary, series and episodes on Disney Plus generally receive higher and more consistent ratings compared to movies. The wider spread in movie ratings suggests that while some movies are well-received, others fall short of audience expectations.

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How do IMDb ratings vary by release year?



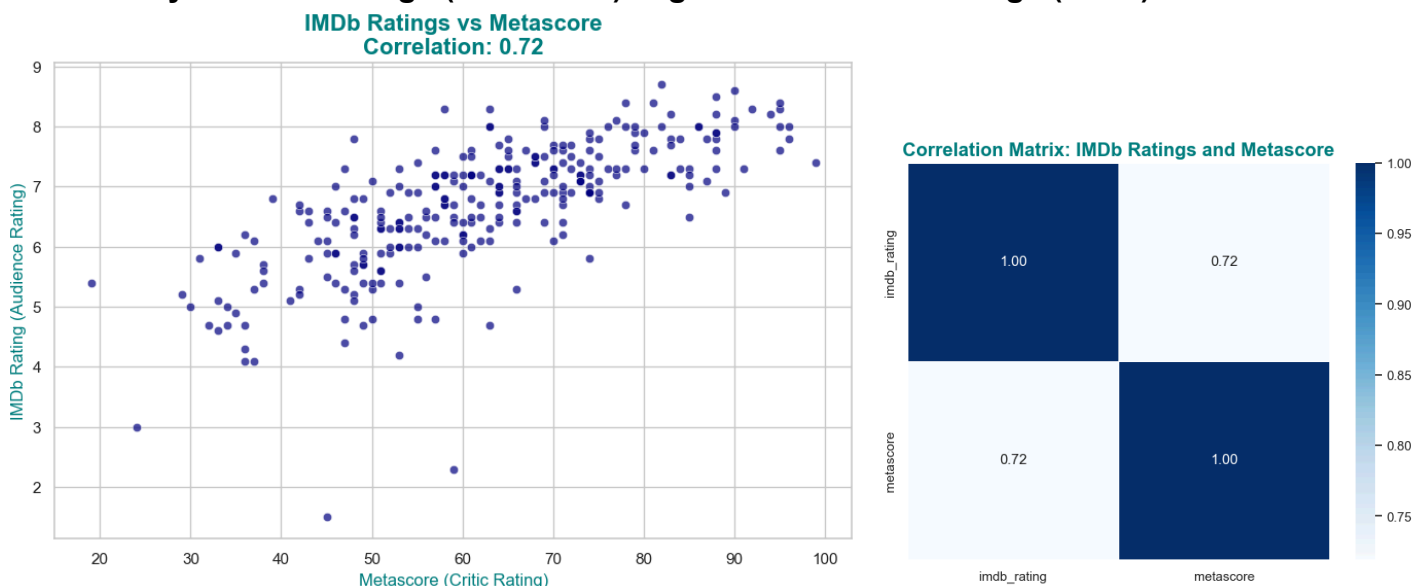
The line plot illustrates how average IMDb ratings for Disney Plus content have varied over time.

In the early years, between 1930 and 1950, ratings generally ranged between 6.5 and 7.7, with notable peaks in 1937 (7.52) and 1947 (7.65), reflecting the positive reception of Disney's early animated classics and storytelling techniques.

During the mid-century period from 1960 to 1980, the ratings displayed more variability, fluctuating between 5.9 and 7.3. The spike in 1980 is largely driven by the well-received *Star Wars: Episode V - The Empire Strikes Back* (8.70 rating).

In recent years, between 2010 and 2020, the ratings show a relatively stable trend, averaging between 6.5 and 7.1. Years like 2014 (7.10) and 2019 (7.03) stand out with higher ratings, suggesting that modern content strategies, such as leveraging popular franchises and diverse storytelling, have been well-received. In summary, Disney Plus content has experienced fluctuations in IMDb ratings over the decades, with recent stable ratings reflecting the success of current content strategies.

How closely do critic ratings (Metascore) align with audience ratings (IMDb)?

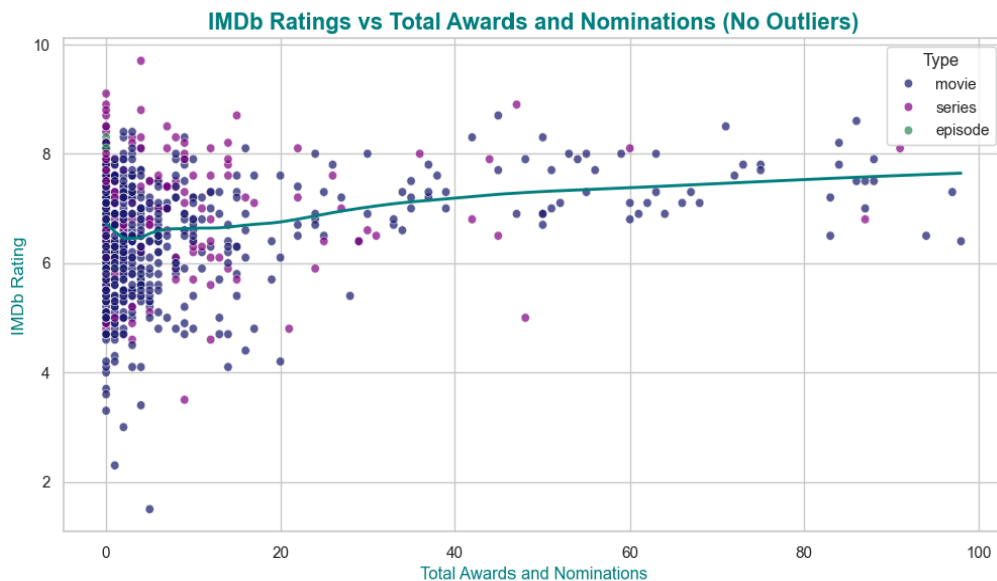


The scatter plot and correlation matrix highlight the relationship between IMDb ratings and Metascore for Disney Plus content. The correlation coefficient of 0.72 indicates a strong positive correlation between these two ratings, meaning that content receiving higher scores from critics also tends to receive higher ratings from audiences. The scatter plot shows a clear upward trend, reinforcing that as Metascores increase, IMDb ratings typically follow a similar pattern. For Metascores between 40 and 70, there is some variability in IMDb ratings, suggesting occasional differences in critic and audience reception.

audience reception for mid-tier content. Content with high Metascores above 80 generally corresponds to IMDb ratings above 7, while content with low Metascores below 40 typically receives IMDb ratings below 5. In summary, critic ratings (Metascore) and audience ratings (IMDb) are closely aligned. This suggests that critically acclaimed Disney Plus content tends to resonate well with audiences, with only slight deviations observed in the mid-range scores.

Awards Analysis

How do the number of awards and nominations correlate with IMDb ratings?



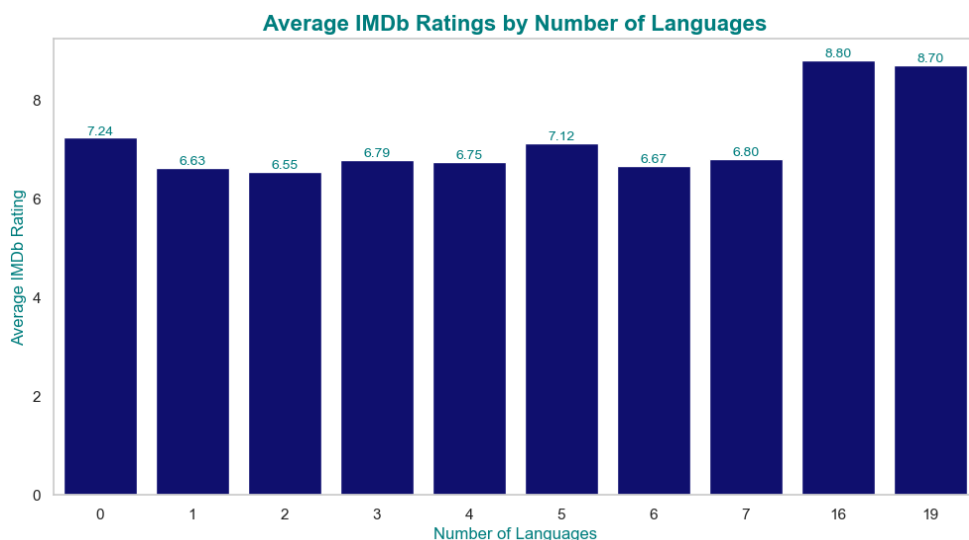
The scatter plot illustrates the relationship between the total number of awards and nominations and IMDb ratings for Disney Plus content, with outliers removed to emphasize the core trends.

Most Disney Plus content has received 10 or fewer awards and nominations, with an average IMDb rating of 6.44. This indicates that the majority of movies and series on the platform are moderately rated and receive limited recognition.

Content that has earned between 11 and 50 awards and nominations shows a slight increase in average IMDb ratings, rising to 6.76. This suggests that content with more critical recognition tends to be rated more favorably by audiences. For titles with 51 to 100 awards and nominations, the average IMDb rating increases further to 7.47, reinforcing the trend that higher recognition correlates with higher ratings. In summary, removing outliers reveals a positive correlation between the number of awards and nominations and IMDb ratings. Disney Plus content that receives more critical recognition generally resonates more positively with audiences, achieving higher ratings.

Diversity Analysis

Do movies or series with more languages tend to have higher IMDb ratings?



The bar plot illustrates the relationship between the number of languages and IMDb ratings for Disney Plus content. Most Disney Plus content is available in a single language, with 721 titles averaging an IMDb rating of 6.63.

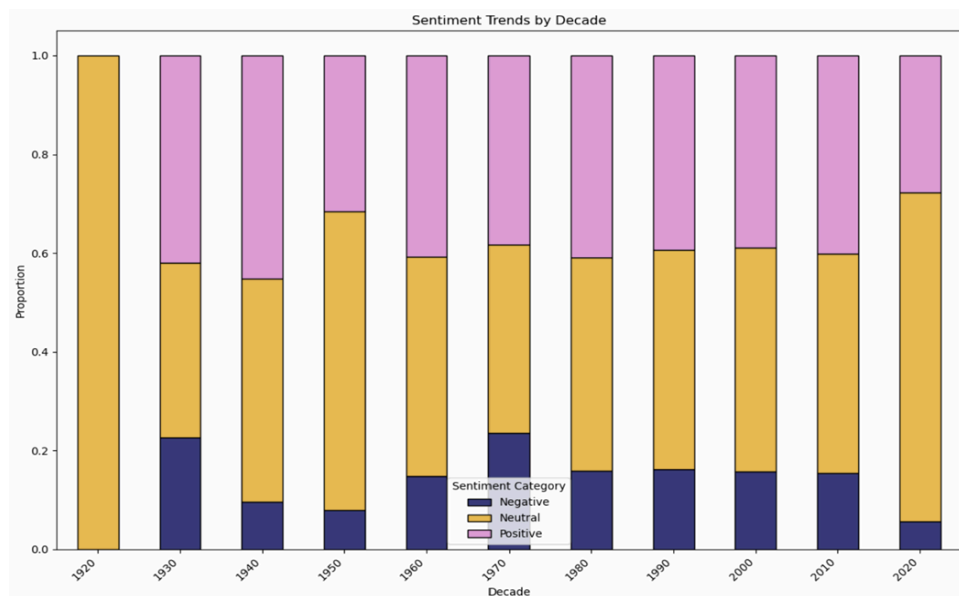
Content available in multiple languages shows slight variation in ratings, but there is no clear trend suggesting that more languages consistently lead to higher ratings.

However, there are notable exceptions: titles available in 16 or 19 languages have exceptionally high ratings of 8.80 and 8.70, respectively, though these cases are rare.

Additionally, 29 titles have 0 languages listed due to missing data. In summary, while highly multilingual content tends to receive higher ratings, the majority of Disney Plus content remains single-language with moderate ratings.

Sentiment Analysis

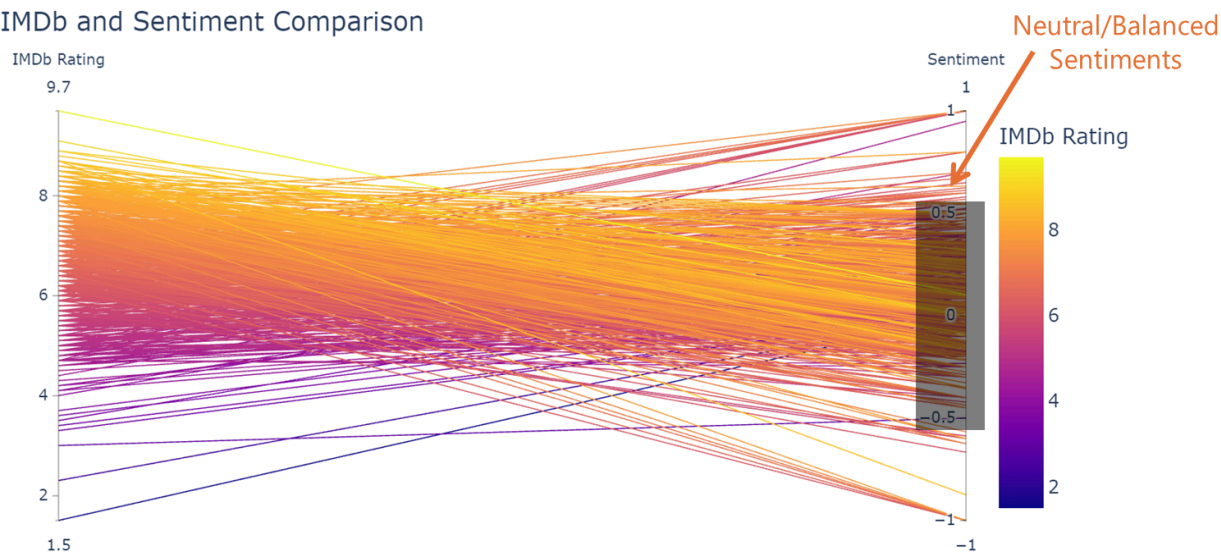
Sentiment Trends by Decade



The bar plot illustrates sentiment trends by decades. The proportion of positive sentiment (pink) increases greatly starting from the 1960s and remains consistently high in the subsequent decades. This trend might reflect societal changes, audience preferences, or a shift toward uplifting storytelling in movies.

How do movie plot sentiments align with IMDb ratings?

IMDb and Sentiment Comparison



The parallel coordinates chart compares IMDb ratings with plot sentiment to explore potential relationships between a movie's emotional tone and its audience ratings. The color gradient represents IMDb ratings, with yellow indicating higher ratings and purple indicating lower ratings. The chart shows that movies with higher IMDb ratings tend to cluster around neutral sentiment values, close to zero which suggests that audiences may prefer movies with balanced emotional tones. Lower rated movies exhibit a wider spread of sentiment, ranging from highly negative (-1) and highly positive (1), indicating that extreme emotional tones, negative or positive are associated with least favorable reception.

Summary of Key Insights

Our analysis of Disney Plus content aimed to uncover strategies that contribute to higher IMDb ratings and greater critical acclaim. Here are the key insights from our exploration:

1. **Awards and Critical Acclaim Align with Higher IMDb Ratings:** Content with a higher number of awards and nominations tends to receive better IMDb ratings. This suggests that critically acclaimed content resonates more positively with audiences.
2. **Animations Consistently Perform Better Than Other Content Types:** Animated content on Disney Plus tends to have higher and more consistent IMDb ratings compared to live-action content. This indicates that Disney's strength in animation continues to appeal to audiences.
3. **Multilingual and Multi-Country Productions Show No Consistent Impact on Ratings:** While there are a few high-rated titles available in many languages or produced across multiple countries, the overall analysis shows no clear relationship between multilingual or multi-country production and higher IMDb ratings. The majority of content is single-language and single-country, with moderate ratings.
4. **Critic and Audience Ratings Exhibit a Strong Correlation:** A correlation coefficient of 0.72 between Metascores (critic ratings) and IMDb ratings (audience ratings) indicates a strong alignment. Content that is highly rated by critics is generally well-received by audiences, demonstrating consistent quality perception across both groups.
5. **Neutral or Balanced Sentiments Align with Higher IMDb Scores:** Content that evokes neutral or balanced sentiment tends to have higher IMDb ratings, suggesting that audiences appreciate content that avoids extremes in tone or emotion.

Conclusion and Recommendations

Disney Plus should prioritize the production of high-quality content with strong potential for awards and critical acclaim, as such productions greatly enhance the platform's reputation and foster a loyal and diverse audience base. Leveraging the success of top-performing genres, which consistently drive higher box office revenue and audience engagement, will align with viewer preferences and secure sustained financial returns.

Additionally, sentiment analysis offers a strategic advantage by providing insights into audience reception, helping to refine storytelling and identify opportunities for improvement. Expanding accessibility through multilingual content and catering to diverse global regions can further extend Disney Plus' international reach and impact.

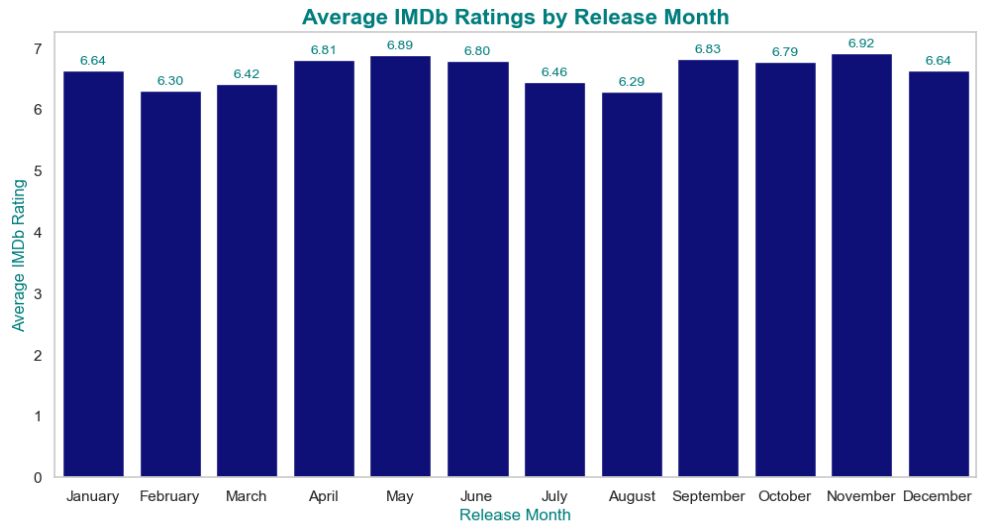
Genres associated with neutral sentiments should remain a focus, ensuring a balanced content portfolio that appeals to a broad range of audience tastes. By combining critical acclaim, genre alignment, and audience insights, Disney Plus can solidify its position as a leading global entertainment platform.

Appendix

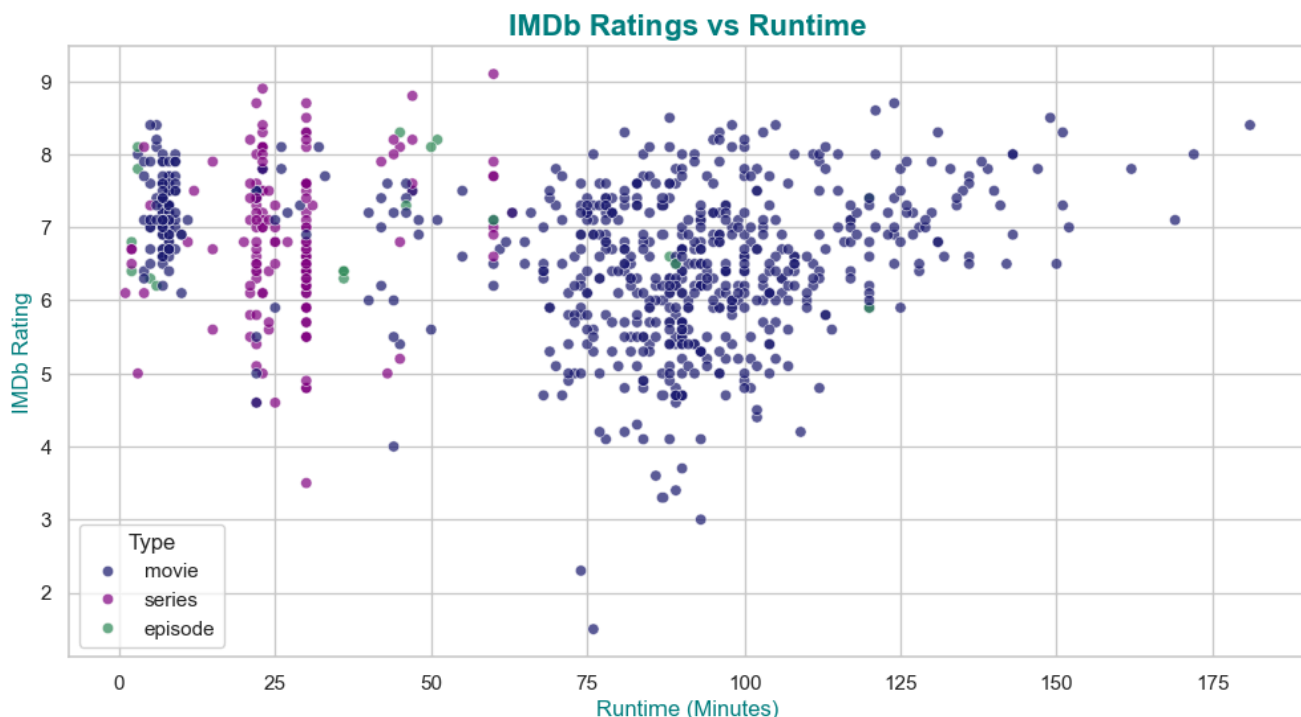
Some analyses were moved to the appendix due to their limited impact on the overall findings. Specifically, the relationship between release month, runtime, and country representations with IMDb ratings showed no meaningful trends. These insights, while informative, revealed consistent or weak correlations that did not provide actionable content strategies for Disney Plus. Including them in the appendix ensures the main report remains focused on the most compelling and relevant discoveries.

Do certain months have higher IMDb ratings for released content?

The bar plot shows the average IMDb ratings for Disney Plus content based on the month of release. The differences in IMDb ratings across months are relatively small, ranging between 6.29 and 6.92. November has the highest average rating at 6.92, while August has the lowest at 6.29. While months like May (6.89) and September (6.83) also show slightly higher ratings, these differences are minimal and do not suggest a strong relationship between the month of release and content quality. In summary, there does not appear to be a relationship between the month of release and IMDb ratings. The ratings remain relatively consistent throughout the year, indicating that other factors, such as content type, genre, or awards, may have a greater influence on ratings.

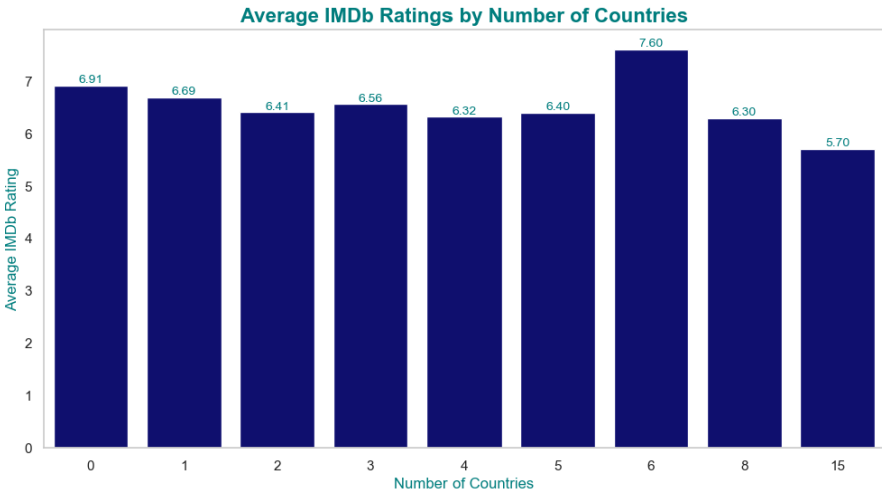


How does the runtime affect IMDb ratings?



The scatter plot illustrates the relationship between runtime and IMDb ratings for Disney Plus movies, series, and episodes. Movies dominate the dataset, with runtimes ranging between 45 and 125 minutes, and their IMDb ratings are spread between 5 and 8, reflecting varied audience reception. Episodes, which typically have shorter runtimes under 30 minutes, cluster around higher ratings between 6 and 8. This suggests that shorter content, like episodes, tends to be well-received. Series, with runtimes between 20 and 50 minutes, show consistent ratings between 5 and 8, similar to movies but with less variability. In summary, shorter content, such as episodes, often receives higher ratings, while movies show more variation in both runtime and ratings. This implies that content strategies focusing on concise, high-quality episodes could contribute to better audience reception.

Do movies or series with more country representations tend to have higher IMDb ratings?



The bar plot displays the relationship between the number of countries involved in production and IMDb ratings for Disney Plus content. Most titles are produced in a single country (732 movies and series), with an average IMDb rating of 6.69. Content produced across two countries (85 titles) shows a slightly lower average rating of 6.41. Titles produced in three to five countries (26 entries) maintain moderate ratings between 6.32 and 6.56. There are exceptions among the outliers; one title produced in eight

countries has a rating of 6.30, while another produced in 15 countries has a lower rating of 5.70. Interestingly, a title produced across six countries stands out with a higher average rating of 7.60. In summary, there is no consistent relationship between the number of countries involved in production and IMDb ratings. While multi-country productions can achieve higher ratings, most Disney Plus content is produced in a single country and receives moderate ratings.

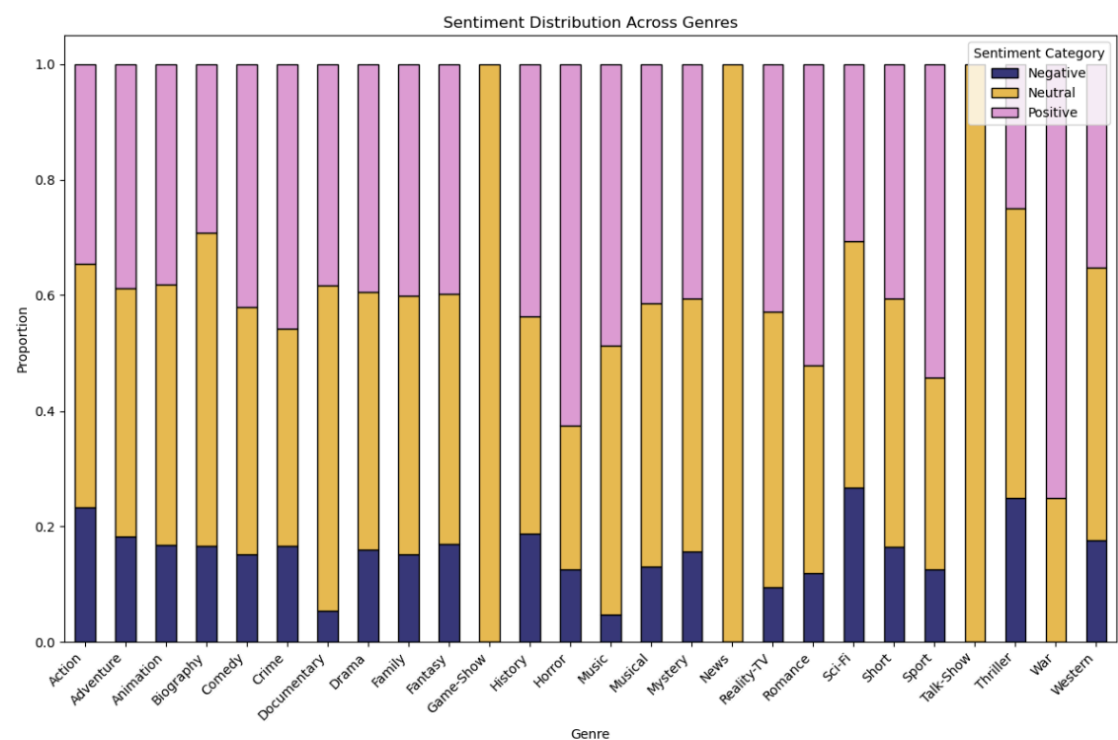
Which Disney Plus movies or series have received the most awards, and how do their IMDb ratings compare?

	title	type	total_awards_nominations	imdb_rating
875	The Simpsons	series	471	8.70
487	Black Panther	movie	376	7.30
40	Avatar	movie	218	7.80
382	Inside Out	movie	212	8.20
720	Star Wars: Episode VII - The Force Awakens	movie	192	7.90
956	WALL·E	movie	185	8.40
486	Avengers: Endgame	movie	167	8.40
950	Up	movie	162	8.20
925	Toy Story 3	movie	157	8.30
493	Guardians of the Galaxy	movie	154	8.00

The table showcases Disney Plus movies and series with the highest number of awards and nominations. *The Simpsons* leads with 471 awards and nominations and an impressive IMDb rating of 8.7. Critically acclaimed animated films such as *WALL·E* and *Inside Out* are also highly recognized, with IMDb ratings exceeding 8.0. Blockbusters like *Avatar* and *Avengers: Endgame* feature prominently on this list, reflecting both high award counts and strong IMDb ratings. These titles demonstrate that both animated classics

and major franchise films contribute to Disney Plus's critically acclaimed content library.

Sentiment Distribution Across Genres



Across most genres, neutral sentiment (yellow) is the most common, highlighting that balanced emotional tones dominate storytelling regardless of genre.